

Arundhati Vasisht

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EDUCATION

University of Massachusetts, Amherst

BS, Computer Science; GPA: 3.83; Chancellor's Award, Dean's List Honoree

Amherst, MA

Expected Fall 2028

SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Cloud: React.js, Node.js, Express, Flask, TensorFlow/Keras, AWS (Bedrock, Lambda, DynamoDB, OpenSearch), MongoDB, PostgreSQL

AI & Tools: RAG, LLM integration, Qdrant, MediaPipe, Grok, XRPL, Solana/Metaplex, REST APIs, Git

EXPERIENCE

Massachusetts Executive Office of Health & Human Services (EOHHS)

Boston, MA

AI Software Engineer

Spring to Summer 2026

- Built an AI-powered regulatory chatbot for Massachusetts DPH's statewide Metrik platform, targeting 85%+ accuracy across 351 local health departments and 5 inspection domains.
- Architected an AWS Bedrock RAG pipeline grounded in official Massachusetts CMR documents, delivering near-instant regulatory answers while reducing inspectors' 2–10 minute manual document searches.
- Designed statewide audit logging, inspector feedback, and response-flagging workflows to track accuracy, completeness, citations, and retrieval quality across inspector interactions.

Indian Institute of Science (IISc)

Bangalore, India

Machine Learning Research Intern

Summer 2025

- Evaluated 6 ML/DL pipelines across 8,700+ chest X-ray and brain MRI images, implementing SVM, CNN, XGBoost, PCA hybrids, and VGG16 transfer learning for medical image classification.
- Achieved up to 96% test accuracy on brain MRI and 95% on chest X-rays; evaluated accuracy, F1, ROC-AUC, and log-loss across multiple splits while collaborating with 11 researchers.

Minutemen Alternative Investment Fund (MAIF)

Amherst, MA

Quant Arm, Junior Analyst

Fall 2025 to Present

- Built a multi-agent NLP pipeline matching prediction-market contracts across 4 exchanges (Kalshi, Polymarket, Manifold, Metaculus) to identify deterministic arbitrage opportunities.
- Embedded market questions in Qdrant, clustered candidates via KNN, and used Groq/Llama to classify 5 semantic relationships; built live quote polling and alerts for configurable profit thresholds.

PROJECTS

KnowYork | Autonomous AI Agent & Blockchain Platform

Fall 2026

- Built a full-stack AI agent that verifies real-world NYC visits and autonomously executes RLUSD rewards across XRPL and Solana, combining Grok vision + multi-sample GPS verification to achieve 90%+ validation accuracy with <4s median decision latency.
- Engineered a zero-trust transaction pipeline with dual-agent verification, deterministic payout controls, and isolated wallet keys, rejecting 95%+ of simulated fraudulent submissions while enforcing a 10 RLUSD on-chain maximum exposure across 25 adversarial security tests.

MedEcho | AI-Powered Medical Results Translator

Spring 2026

- Built a React/Flask system using DenseNet-121 and GradCAM to classify chest X-rays, highlight diagnostic regions, and translate findings into patient-friendly explanations across 4 languages.
- Integrated Gemini, ElevenLabs, and HeyGen LiveAvatar for multilingual result delivery and the ClinicalTrials.gov API for patient-trial eligibility matching; built the complete prototype in 24 hours.

FingerSolve | AI-Powered Hand Gesture Recognition Platform

Spring 2025

- Built a real-time React/Flask gesture-recognition platform for touch-free math quizzes, improving engagement by 40% and reducing user input errors by 30% during pilot testing.
- Engineered MediaPipe Hands + TensorFlow/Keras classification across 12 gestures with 94% accuracy; deployed a prototype that earned 2nd place at the MassAI ML Conference and attracted interest from 50+ educators.

EcoForecast | Full-Stack Sustainability Analytics Platform

Spring 2026

- Built a full-stack React/TypeScript platform simulating 20-year financial ROI and carbon impact of green investments, with Node.js APIs and MongoDB workflows processing quarterly utility data.
- Developed cost, break-even, and carbon-reduction models with 4+ interactive visualizations and integrated a LLaMA 3.3-powered AI advisor for context-aware sustainability recommendations.

LEADERSHIP & INVOLVEMENT

HackUMass

Amherst, MA

Organizer, Technical & Public Relations Team

Fall 2025 to Present

- Develop and maintain HackUMass website and participant dashboard features using React, Node.js, Express, and PostgreSQL while supporting technical operations and public relations for the annual hackathon.

Undergraduate Course Assistant, COMPSCI 198C | Manning College of Information and Computer Sciences, UMass Amherst

Spring 2026